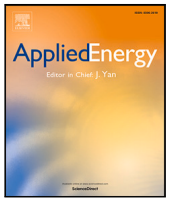
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# An IoT-based energy management system for AC microgrids with grid and security constraints

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ABSTRACT

This paper proposes an Internet-of-Things (IoT) based energy management system (EMS) for the optimal operation of unbalanced three-phase AC microgrids. The system utilizes a software architecture based on microservices, which includes a stochastic economic dispatch optimizer (EDO), a database, a web-based graphical user interface (GUI), and an application programming interface (API). The EDO uses a mixed-integer linear programming (MILP) model to ensure the day-ahead dispatch of the distributed energy resources (DERs) in the microgrid while adhering to grid constraints such as voltage, current, and power limits. Additionally, the optimization module takes into account security constraints for unplanned islanded operation, as well as stochastic scenarios of local demand and renewable generation. To assess the performance of the proposed IoT-based EMS, tests are conducted using a real-time simulator in a software-in-the-loop (SIL) experimental setup. Actual data from a microgrid located at the State University of Campinas (UNICAMP) in Brazil is utilized for the tests. The microgrid consisted of a photovoltaic (PV) system, a battery energy storage system (BESS), a thermal generation unit, and variable demands. Results indicated the effectiveness of the proposed IoT-based EMS in monitoring the operation of the microgrid and defining the optimal day-ahead dispatch of local DERs.

## 1. Introduction

The integration of various distributed energy resources (DERs) such as battery energy storage systems (BESSs), photovoltaic (PV) systems, wind generators, electric vehicle (EV) chargers, and thermal generation units has presented new difficulties for monitoring and operating electrical distribution systems (EDSs) [1]. Microgrids offer a reliable solution for integrating these DERs and managing controllable and uncontrollable loads through an energy management system (EMS) [2].

The purpose of an EMS is to optimize the operation of DERs within a microgrid, using criteria such as cost minimization, improved grid reliability, emission reduction, and peak shaving [3]. The EMS primarily plans the day-ahead operation of local DERs and may also provide ancillary services such as voltage and frequency regulation, improved reliability, and loss reduction [4,5].

On the other hand, the Internet of Things (IoT) is a modern paradigm of computer networks with numerous applications in the industrial and technology sectors. In smart grids, IoT facilitates communication among agents, making it a useful tool for decision-making [6]. The use of IoT in an EMS drives the growth of microgrids by enabling secure remote monitoring and control of internal DERs [7,8]. For instance, as demonstrated in [9], an EMS for microgrids can be created with the integration of deer hunting optimization and crow search algorithm, to reduce electricity costs and increase profits during grid-connected operations. However, the model does not take into account uncertainties in demand and renewable generation.

In [10], a two-stage framework for scheduling microgrids and reconfiguring a distribution feeder is proposed, considering the uncertainties associated with demand, market prices, and renewable energy sources (RESs). Moreover, an EMS is presented in [11] for optimal microgrid planning, considering uncertainties in the IoT framework, using Benders' decomposition and a genetic-fuzzy algorithm. Although the IoT frameworks presented in these papers show different microgrid elements, considering the grid-connected [9,10] and islanded operation mode [11], they do not implement tests with real-time simulators or cyber–physical systems.

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**Nomenclature**

**Abbreviations**

| | |
|---|---|
| AC | Alternating current |
| API | Application programming interface |
| BESS | Battery energy storage system |
| DC | Direct current |
| DER | Distributed energy resource |
| EDO | Economic dispatch optimizer |
| EMS | Energy management system |
| EV | Electric vehicle |
| FPGA | Field-programmable gate array |
| GUI | Graphical user interface |
| HIL | Hardware-in-the-loop |
| HTTP | Hypertext transfer protocol |
| IoT | Internet-of-Things |
| MILP | Mixed-integer linear programming |
| MINLP | Mixed-integer nonlinear programming |
| OPF | Optimal power flow |
| PCC | Point of common coupling |
| PV | Photovoltaic |
| RES | Renewable energy source |
| REST | Representational state transfer |
| SIL | Software-in-the-Loop |
| SOC | State of charge |

For practical use, an EMS for microgrids must be evaluated under realistic conditions that consider communication, optimization, and stability and allow for human interaction with the system. Few studies have created cyber–physical testbeds in real-time simulation environments. [12] proposes a web-based framework for a system-level EMS in residential AC/DC microgrids and validates the system using a hardware-in-the-loop (HIL) setup in a laboratory. [13] presents an IoT-based optimal demand-side management for controlling a laboratory microgrid. [14] implements an EMS for a hybrid microgrid with PV systems, biogas, and BESS using an IoT-based smart communication platform. However, these studies lack a graphical user interface (GUI) or application programming interface (API), resulting in limited human interaction.

Both the GUI and API offer an interactive, web-based platform, allowing users to exchange data with the microgrid, often in real-time. Furthermore, the electrical distribution grids are commonly represented by a three-phase connection. Depending of the grid characteristics, many single-phase, two-phase or three-phase loads can be connected to the system with an asymmetrical configuration of the conductors in the distribution lines leading to an unbalanced grid. Those aspects are usually disregarded by single-phase systems resulting in unrealistic representations of the electrical grid. A three-phase representation of the system has several advantages over single-phase systems, including increased efficiency, improved power quality, increased capacity, and improved reliability. Therefore, this method is commonly used to model and analyze electrical distribution grids, as it provides a more accurate representation of the system and allows for more efficient analysis and optimization of the grid. A practical implementation for the EMS of an existing microgrid in the Canadian Renewable Energy Laboratory is presented in [2], where a GUI was implemented to ease the operation and display the measurements of the microgrid. However, the optimal power flow of unbalanced three-phase AC systems is not taken into account despite considering power balance. In [7], an IoT environment is integrated with a proposed EMS for data storage, analysis, and remote monitoring. A web-based GUI is developed to display real-time energy consumption and other electrical parameters of a prototype developed in a laboratory environment to emulate a smart home. The tool in [7] focuses on solutions for smart homes and does not account for grid constraints or uncertainties.

Of course, there are numerous EMS commercial solutions for microgrids, such as ETAP $\mu$Grid [15], EcoStruxure Microgrid Operation [16] and Advisor [17], Advanced EMS [18], Spectrum Power MGMS [19], and e-mesh EMS [20]. ETAP $\mu$Grid is a software and hardware platform that provides grid management functions: load shedding and demand management, integrated logic for DER dispatch, transition to islanded mode and back, black-start, load and generation forecasting, and tools to monitor the operation of the system [15]. Schneider Electric presents several solutions to operate, control, and monitor microgrids [21]. EcoStruxure Microgrid Operation manages the energy production, including renewable generation and demand, and EcoStruxure Microgrid Advisor enables control of on-site energy resources and loads. EcoStruxure Microgrid Operation realizes the automatic network connection management, controls the DERs, presents pre-configured solutions, and black-start [16]. The EcoStruxure Microgrid Advisor works with the operation, providing intelligence and optimization to system operation [16]. The Advanced EMS [18], developed by General Electric, is a framework that integrates with distribution utility functions, such as planning and field management, asset and maintenance management, security services, power plant control systems, and others. Hence, its functionalities include automatic generation control, DERs dispatch and curtailment, transaction scheduling, optimal power flow, loss minimization, security constraint dispatch, short-circuit analysis, and other functions. The Siemens Spectrum Power Microgrid Management System [19] is an advanced control and optimization software used to maximize renewable energy resources in coordination with the local utility or wholesale market rates. This software can forecast loads and market prices to find the optimal economic schedules for the day-ahead operation, reducing energy costs and emissions. Furthermore, the optimal coordination of DERs includes islanding, black-start, and grid re-synchronization, among others. Likewise, e-mesh EMS [20] is a software that performs the energy management of DERs and can be extended to integrate renewable, conventional power generation sources, and controllable loads, such as EV chargers. It provides forecasting of loads, renewable generation, and market prices. Although commercial solutions have multiple functionalities and extensive user support, in most cases they are black-boxes. Hence, in the publicly available documentation, no details are given regarding the optimization approach used to deploy the EMS. Beyond that, commercial solutions for microgrids are mostly energy-oriented, meaning that they disregard the internal network's unbalanced three-phase AC nature. Moreover, even though most commercial solutions have forecasting modules, it is not always the case that forecasting errors are incorporated into the optimization process via stochastic or robust approaches.

Alternatively, Open Source Energy Management System (OpenEMS), developed by OpenEMS Association, is a modular platform for monitoring, controlling, and integrating energy storage with RESs, EV charging stations, time-of-use electricity tariffs, among others [22]. The OpenEMS IoT contains three main components: Edge, which communicates with devices and services, collects data, and executes control algorithms; user interface (UI); and Back-end which connects the decentralized Edge systems and provides aggregation, monitoring, and control via IoT [23]. Even though OpenEMS is open-source software (i.e., not a black-box), it disregards the three-phase AC network modeling of microgrids. Table 1 compares the features of the reviewed EMS solutions with the one presented in this paper.

This paper presents the development and implementation of an IoT-based Energy Management System (EMS) for microgrids, aimed at addressing the identified gaps. A stochastic mixed-integer linear programming (MILP) model has been used to formulate the day-ahead optimization module, a.k.a., the economic dispatch optimizer (EDO). The proposed EMS adopts a microservice architecture and is equipped

**Table 1**
Comparison of the proposed method with previous works.

| Ref | Solution method of the optimization | Contingency constraints: Grid-connected | Islanded | Transitions | Uncertainty | 3-phase Microgrid model | Case study | Implemented with | Real-time | GUI |
|---|---|---|---|---|---|---|---|---|---|---|
| [9] | DHO and CSA | ✓ | ✗ | ✗ | ✗ | ✗ | Simulation | MATLAB/ Simulink | ✗ | ✗ |
| [10] | MILP | ✓ | ✗ | ✗ | ✓ | ✗ | Simulation | GAMS | ✗ | ✗ |
| [11] | Benders and GA-Fuzzy | ✓ | ✓ | ✗ | ✓ | ✗ | Simulation | | ✗ | ✗ |
| [12] | DL-NILM | ✓ | ✗ | ✗ | ✗ | Some Comp. | Exp. Setup | MATLAB and GAMS | ✓ | ✗ |
| [13] | GA | ✓ | ✓ | ✗ | ✗ | ✗ | Exp. Setup | MATLAB/ Simulink | ✓ | ✗ |
| [14] | Rule-based | ✓ | ✓ | ✓ | ✗ | Some Comp. | Exp. Setup | PSCAD | ✓ | ✗ |
| [2] | RB and OB | ✓ | ✓ | ✓ | ✗ | ✗ | Testbed | Python | ✓ | ✓ |
| [7] | RB and OB | ✓ | ✗ | ✗ | ✗ | ✗ | Exp. Setup | XBee modules | ✓ | ✓ |
| [15] | Not informed | ✓ | ✓ | ✓ | ✓ | Not informed | NA | Not informed | ✓ | ✓ |
| [16] | Not informed | ✓ | ✓ | ✓ | ✗ | Not informed | NA | Not informed | ✓ | ✓ |
| [17] | Not informed | ✓ | ✓ | ✓ | ✓ | Not informed | NA | Not informed | ✓ | ✓ |
| [18] | Not informed | ✓ | ✓ | ✓ | ✓ | Not informed | NA | Not informed | ✓ | ✓ |
| [19] | Not informed | ✓ | ✓ | ✓ | ✓ | Not informed | NA | Not informed | ✓ | ✓ |
| [20] | Not informed | ✓ | ✓ | ✓ | ✓ | Not informed | NA | Not informed | ✓ | ✓ |
| [23] | Not informed | ✓ | ✓ | ✓ | ✓ | Not informed | NA | Java and Angular | ✓ | ✓ |
| This paper | MILP | ✓ | ✓ | ✓ | ✓ | ✓ | Exp. Setup | Python and Angular | ✓ | ✓ |

* DHO and CSA — Deer Hunting Optimization and Crow Search Algorithm, GA — Genetic Algorithm, DL-NILM — Deep Learning-Non-Intrusive Load Monitoring
* RB and OB — Rule-based and Optimization-based EMS, PSO — Particle Swarm Optimization, AOFL — Adaptive Optimal Fuzzy Logic, NA — Not applicable.

with a database, a web-based GUI, and an API, for complete system deployment and integration. The EDO determines the optimal day-ahead dispatch of the available DERs through an optimal power flow calculation for unbalanced three-phase AC networks, while taking into account grid constraints (such as three-phase voltage, current, and power flow magnitude limits) and security constraints (such as the possibility of unintentional disconnection from the main grid). Additionally, the EDO allows for direct control of loads and PV systems, independently. The IoT-based EMS enables users to configure hourly energy prices, thermal generation costs, and load curtailment costs, to guide the EDO policies. Tests are conducted using a real-time simulator in a software-in-the-loop (SIL) experimental setup, implemented on a field-programmable gate array (FPGA) using Typhoon HIL [24]. Real data from a microgrid located at the State University of Campinas (UNICAMP), in São Paulo, Brazil, is used. The microgrid includes a PV system, a BESS, a thermal generation unit, and variable demands.

Based on the identified gaps, this paper presents the development and implementation of an IoT-based EMS for microgrids. A stochastic mixed-integer linear programming (MILP) model formulates the day-ahead optimization module, a.k.a., the economic dispatch optimizer (EDO). Furthermore, the proposed EMS is based on a microservice architecture, equipped with a database, a web-based GUI, and an API, for complete system deployment and integration. The EDO defines the optimal day-ahead dispatch of the available DERs via an optimal power flow for unbalanced three-phase AC networks, considering grid constraints (i.e., three-phase voltage, current, and power flow magnitude limits) and security constraints (i.e., the EDO considers that an unintentional disconnection from the main grid could occur at any given moment). Additionally, the EDO allows the direct control of loads and PV systems, independently. The proposed IoT-based EMS allows users to configure hourly energy prices, thermal generation costs, and load curtailment costs to guide the EDO policies. Tests are conducted using a real-time simulator in a software-in-the-loop (SIL) experimental setup implemented on a field-programmable gate array (FPGA) using Typhoon HIL [24]. Real data from a microgrid located at the State University of Campinas (UNICAMP), in São Paulo, Brazil, is used. The microgrid has a PV system, a BESS, a thermal generation unit, and variable demands. The main contributions of this work are:

- Development of a complete IoT-based EMS for microgrids that incorporates a database, a web-based GUI, an API, and a stochastic EDO to determine the optimal day-ahead dispatch of DERs through mathematical programming, while considering grid and security constraints.
- Validation through real-time simulation in a SIL experimental setup, considering microgrid operation in grid-connected and islanded modes, and transitions between the two modes.
- Definition of a default day-ahead operation for cases in which the EDO is unable to find the optimal solution within a given time, or when updated information from the microgrid is unavailable (e.g. due to communication issues).

## 2. Proposed IoT-based EMS

The proposed IoT-based EMS platform for microgrids comprises five modules: a stochastic EDO module, an API, a database, a job scheduler commonly known as Cron, and a web-based GUI. Fig. 1 shows the general architecture of the proposed IoT-based EMS. The application is developed using digital containers in order to make it scalable, agnostic to any operative system, and easy to incorporate into a microservice architecture [25]. Containers are built in Docker [26] and orchestrated using Docker-Compose [27]. Four Docker containers are created: `Python-Docker-Container`, which compiles and executes the EDO, the API, and the Crons, all developed in Python [28]; `MySQL-Docker-Container`, a relational database to persist historical information and system data; `Nginx-Docker-Container`, an open-source web server called Nginx [29] used for request management; and `Angular-Docker-Container`, which contains and runs the web-based GUI. In this case, Docker-Compose defines and runs the Docker containers as services [27]. More details about each container will be provided in the following subsections.

As shown in Fig. 1, the IoT-based EMS communicates with the microgrid internal controllers through an external API. In this case, a dedicated API has been developed (called SIL-API), to allow easy interaction between external resources and the internal controllers of the microgrid. Thus, a RESTful [30] link between the proposed IoT-based EMS and the microgrid resources is created, which allows a secure exchange of information and control actions through the Internet. The Cron module of the EMS makes HTTP requests to the SIL-API every five minutes, collecting microgrid operation data. On the other hand, every twenty-four hours, the Cron triggers the EDO module to define a new day-ahead dispatch. Finally, this new dispatch is also sent to the internal microgrid controllers via HTTP requests. Microgrid network data, parameters of the EDO, and DER dispatches are stored in the database. The real-time monitoring of the microgrid and the economic day-ahead dispatch can be visualized and parameterized through the frontend, a.k.a., the web-based GUI. Nginx is responsible for processing all HTTP requests coming from the SIL-API, the frontend, and the API; making sure that data traffic is efficiently balanced and secured.

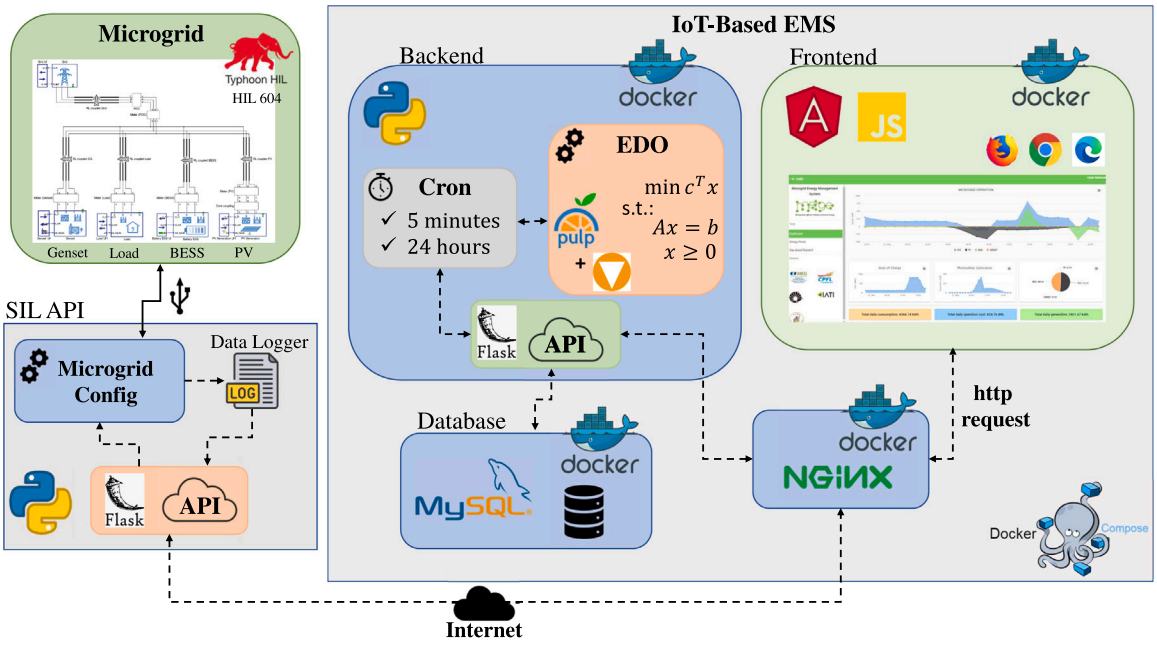



**Fig. 1.** General architecture of the proposed IoT-based EMS.

### 2.1. SIL-API

A microgrid was modeled in Typhoon HIL [24], which enables the implementation and validation of microgrids with real-time controllers. Typhoon HIL has a collection of functions that allows users to control the HIL simulation process from Python scripts [31]. Hence, the SIL-API module was developed to integrate the IoT-based EMS with the microgrid simulated in Typhoon HIL. The physical communication of the SIL-API with the FPGA that runs the simulation is realized via USB protocol. This module has three components as shown in Fig. 1: Microgrid Config, Data Logger, and an API.

The Microgrid Config component starts and controls the simulation process, it also defines and changes the simulation status based on the dispatch specified by the EDO, and it collects the real-time measurements of the microgrid operation. Data acquisition is automatically performed every minute and stored in a Data Logger. Finally, an API was implemented using Flask [32] to integrate the operation of the Microgrid Config with the proposed IoT-based EMS. This API provides, among other things, the last stored record in the Data Logger.

The representational state transfer (REST) software architectural style, facilitates the communication between different cyber-systems. In this way, the microgrid control interface and the IoT-based EMS are carried out independently, adding reliability and fast performance to the integration of services. Note that the proposed IoT-based EMS can be used with other types of microgrids, either real or simulated, as long as an API that allows communication via REST is available on both sides.

### 2.2. Stochastic economic dispatch optimizer (EDO)

The main objective of the EDO module is to define the optimal day-ahead scheduling of available DERs in the microgrid, i.e., charging and discharging schedules of the BESS, PV curtailment, and/or load curtailment, non-renewable distributed generation (a.k.a., genset) dispatch, as well as, the optimal energy exchange with the main grid.

The AC optimal power flow (OPF) for three-phase unbalanced microgrids can be modeled as a mixed-integer nonlinear programming (MINLP) problem, which through efficient linearization strategies [33,34], can be recast into a mixed-integer linear programming (MILP) problem. The objective function in (1) minimizes the average operating costs of the microgrid for day-ahead scheduling while satisfying the technical and physical constraints of the network. The objective function considers the cost of purchasing electricity from the main grid, the cost of the genset operation, and the cost of load/PV curtailment. More details of the formulation can be found in [35].

$$\min \{\text{Operating Costs} + \text{Cost of Load/PV shedding}\} \tag{1}$$

subject to:

- Three-Phase Unbalanced AC OPF
- Genset operation
- BESS operation
- Security constraints

The three-phase unbalanced AC OPF incorporates constraints that calculate the active and reactive power losses, the active and reactive power balance in the system, calculate the voltage magnitude in each node, establish limits for voltage and current magnitudes, and define the maximum current in the point of common coupling (PCC). The loads and PVs are modeled as constant active and reactive injection components. Thus, the EDO allows selective load curtailment and PV disconnection when necessary. Independent binary decision variables are used to define whether there will be load curtailment or PV system disconnection during the microgrid operation. As mentioned, the load curtailment is penalized with a high cost in the objective function. Constraints related to the operation of the genset define the total active and reactive power dispatches and impose active and reactive power limits.

According to [35], there is a set of constraints that models the operation of the BESS, defining its state of charge (SOC), total input/output power, and the three-phase dis/charging active power. Other equations guarantee the balanced operation and specify limits for the three-phase dis/charging active power and energy capacities. Moreover, an unexpected islanded operation at any given time is also considered. To do so, security constraints are defined, disconnecting the microgrid from the main grid during the given outage [36].

Predictions of stochastic time series require a historical analysis of local data, which introduces uncertainties due to forecasting inaccuracies [37–39]. These uncertainties associated with load and PV forecasts are incorporated as scenarios into the model. Therefore, if the forecast

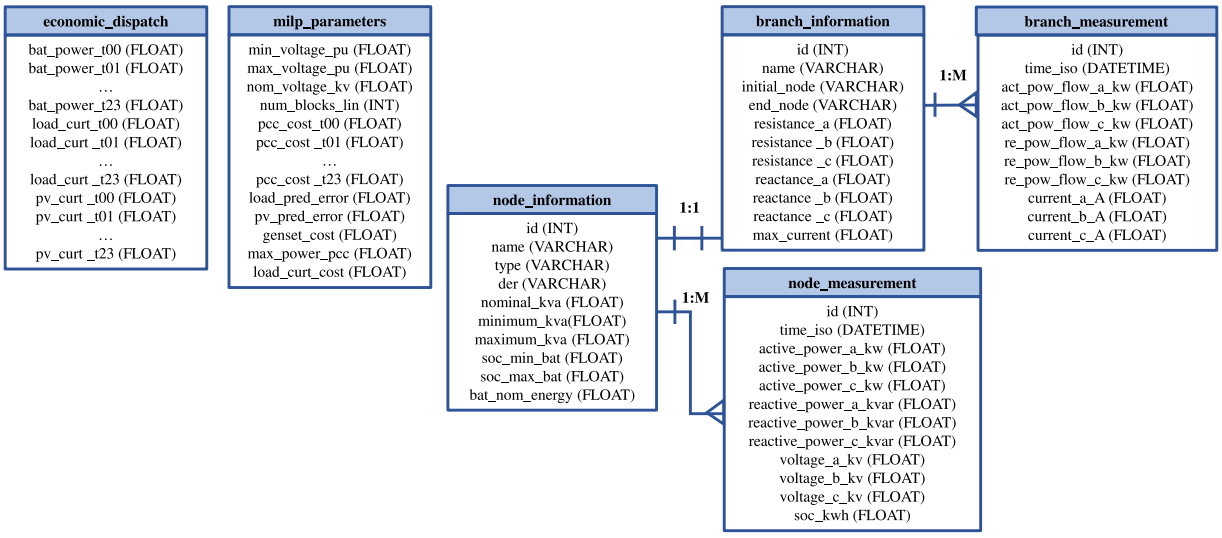



**Fig. 2.** Database structure.

indicates that, in the next 24 h, the solar radiation is very uncertain due to the presence of clouds, then the forecast error will be higher, and the scenarios of the mathematical model will reflect this variance. Unlike [35], the uncertainties related to the reference voltage and the modeling of EV chargers are not considered. As the proposed IoT-based EMS has a microservices framework, forecasting tools can be easily integrated as independent services. Finally, the MILP model was implemented in the mathematical programming language PuLP [40] and solved through the open-source solver Mosek [41].

### 2.3. Database

Measured data during microgrid operation, such as nodal voltages, branch currents, power flows, and power injections, are stored in a database implemented using the MySQL server [42]. MySQL is a popular open-source relational database management system [42]. In MySQL, the user establishes the schema of the database, organizes data into tables based on his/her requirements, and configures rules to control the relationship between tables [43]. Fig. 2 shows the proposed database structure.

The related data has been grouped in six tables: `economic_dispatch` for storing dispatch data of the BESS, load curtailment, PV generation disconnection, and genset dispatch; `milp_parameters`, which contains the input parameters used by the EDO, for example, minimum, maximum and nominal voltage, grid power purchase costs, load curtailment costs, operational cost of the genset, and others; `node_information` for storing information such as node name and type, type of DER, nominal, minimum and maximum apparent power, and the nominal, minimum and maximum SOC for the BESS node; `branch_information` for storing information such as the name of the branch, initial and end node, branch resistance and reactance for each phase, and maximum current; `node_measurement` to store nodal voltages, injections of active and reactive power and SOC for the BESS node; and finally, `branch_measurement` to store active and reactive power flow values, and current in the branches.

Note that the node information table has a one-to-one (1:1) relationship with the branch information table. Hence, each branch information is directly related to each node; and has a one-to-many (1:M) relationship with node measurement because different measurements are performed for each node of the microgrid under study. Additionally, the branch information table has a one-to-many (1:M) relationship with the branch measurement table because several measurements are also performed for each branch.

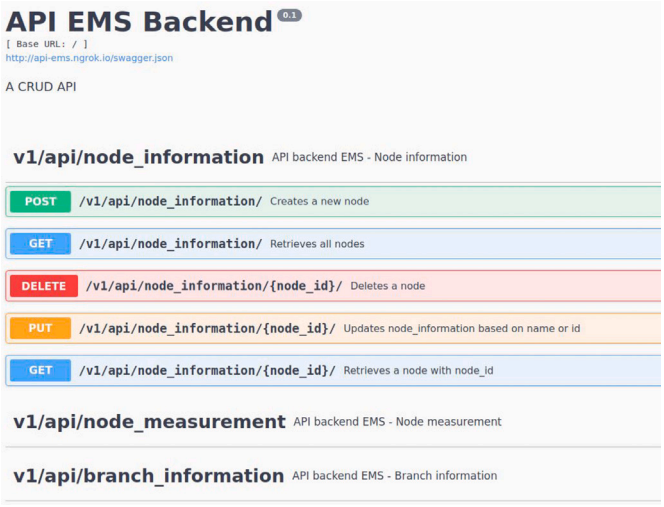



**Fig. 3.** Endpoints of the proposed EMS API.

### 2.4. EMS API

An API is a set of software routines used to integrate applications that exchange information. Hence, the proposed EMS API is responsible for communication and integration among the backend, the database, and the frontend. The API is implemented using Flask [32], a framework for creating web applications in Python. Additionally, the Swagger [44] tool is used as a simple maintenance GUI. Swagger has tools that can immediately test the written APIs, visualizing and interacting with the API's resources, in a friendly manner [45].

Fig. 3 shows the endpoints, e.g., `/v1/api/node_information/` and others, and the methods indicating the allowed interactions with the EMS API, such as GET, POST, PUT, and DELETE.

### 2.5. Crons

The proposed IoT-based EMS communicates with the microgrid controllers through an external API. Measurements (i.e., voltage and current magnitudes, power flows, etc.) are collected via HTTP requests from the SIL-API every five minutes. On the other hand, the day-ahead

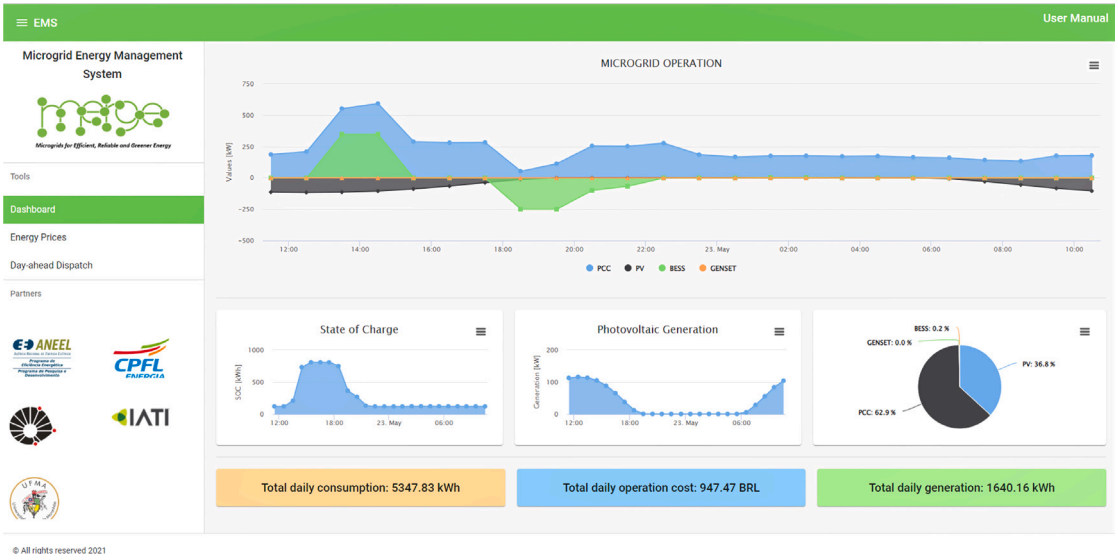



**Fig. 4.** Dashboard of the IoT-based EMS.

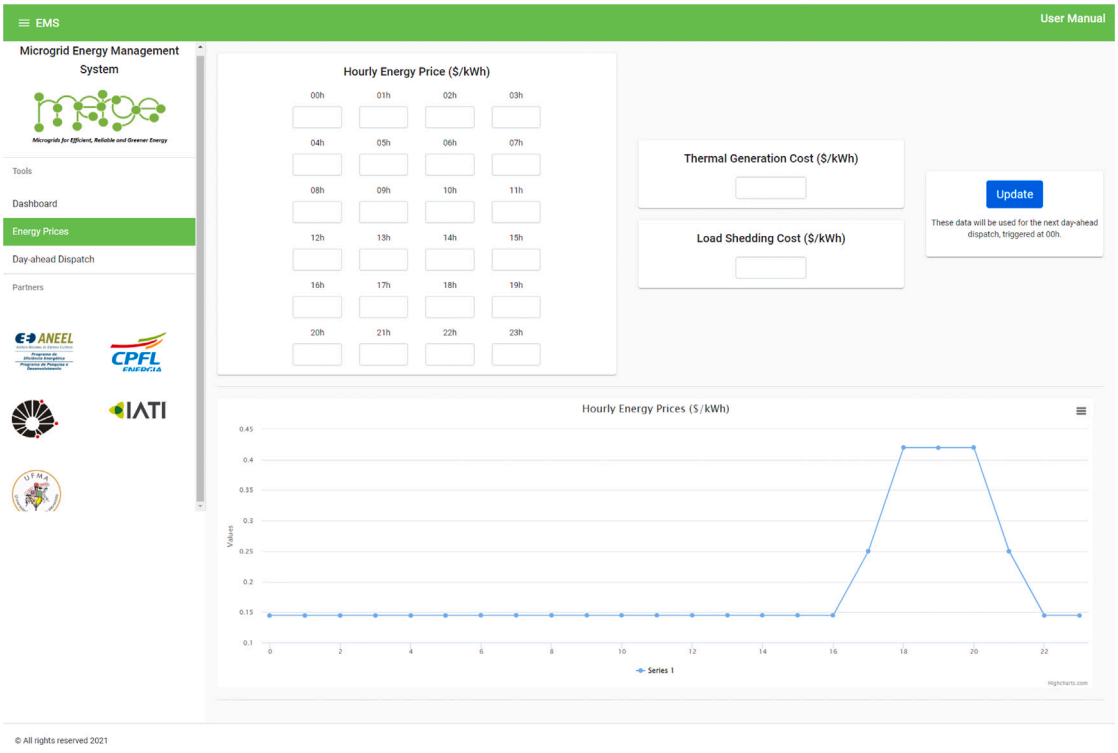



**Fig. 5.** Energy Prices tab of the EMS.

dispatch of the DERs is defined once, at the beginning of each day. Therefore, two Crons were implemented to execute these two tasks automatically. If the microgrid is online and operational, one Cron performs HTTP requests to the SIL-API every five minutes. Another Cron is performed every 24 h, which triggers the stochastic EDO module to define a new day-ahead dispatch.

### 2.6. Graphical user interface (GUI)

A friendly GUI allows the user to browse the web application interactively, view graphics, and configure internal settings. The frontend was implemented using Angular [46], open-source software that uses Typescript as its main programming language. The implemented GUI has a single-page environment, suitable for any browser, and it can be accessed from any device that has access to the IoT-based EMS.

Fig. 4 presents the main dashboard of the GUI. The structure of the web app is divided into three parts: (1) a header, which corresponds to the top menu bar in green color, with a "User Manual" button that allows the user to download the document with guidelines for installing and using the EMS; (2) a footer, corresponding to the bottom bar that contains information about the current version of the app; and (3) a sidebar, which allows the user to select one of the three following environments:

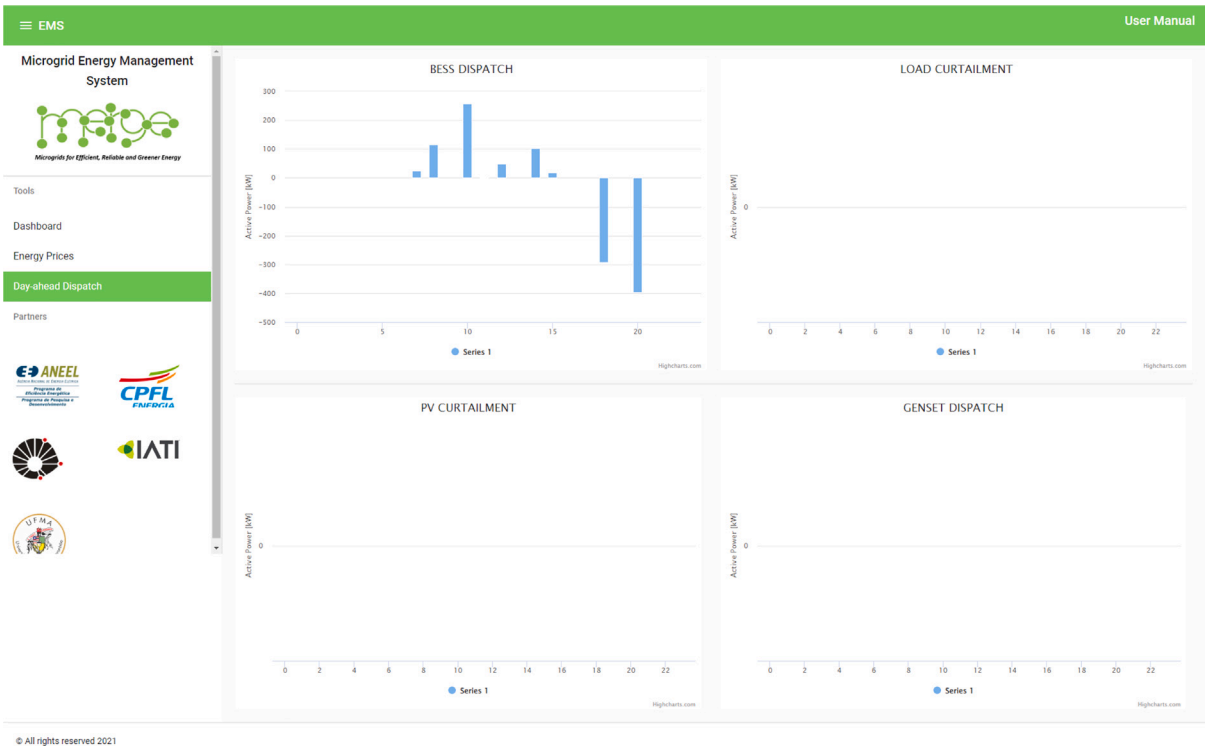



**Fig. 6.** Day-ahead Dispatch tab of the EMS.

### 2.6.1. Dashboard

The dashboard is shown on the home screen when accessing the GUI for the first time (see Fig. 4). It is possible to visualize graphs with the operation of the microgrid in real-time, the SOC of the BESS, the active power of the PV generation system, and a pie chart that compares the energy sources used in the microgrid to supply the demand in real-time. The total daily consumption, operation cost, and generation are displayed at the bottom of the page.

### 2.6.2. Energy prices

Fig. 5 shows the Energy Prices tab of the IoT-based EMS. In this tab, the user can configure the hourly energy prices of the main network, the costs related to thermal generation (genset) operation, and load shedding costs. After submission, the prices are sent to the database, and a graph will be displayed below. Note that these data will be used for the next day-ahead dispatch, triggered at 00 h.

### 2.6.3. Day-ahead dispatch

The Day-ahead Dispatch module (see Fig. 6) displays the defined dispatch by the EDO. It shows the dispatch of the BESS (positive values represent recharge and negative values represent discharge), the load curtailment, the PV power curtailment, and the genset dispatch. For the particular case shown in Fig. 6, the EDO chose not to dispatch the genset, and it did not perform any load shedding or PV curtailment.

## 3. EMS algorithms

This section outlines the algorithms utilized in the proposed IoT-based EMS. The data acquisition and processing for the microgrid is depicted in the flowchart in Fig. 7. When the system is initiated, the EMS sends an HTTP request to the SIL-API every 5 min to retrieve the latest microgrid measurements. If the data cannot be obtained due to the SIL-API being disconnected, a warning 301 (refer to Table 2) is displayed. If the data is successfully obtained, a warning 302 is displayed, and the measured data is displayed in the frontend.

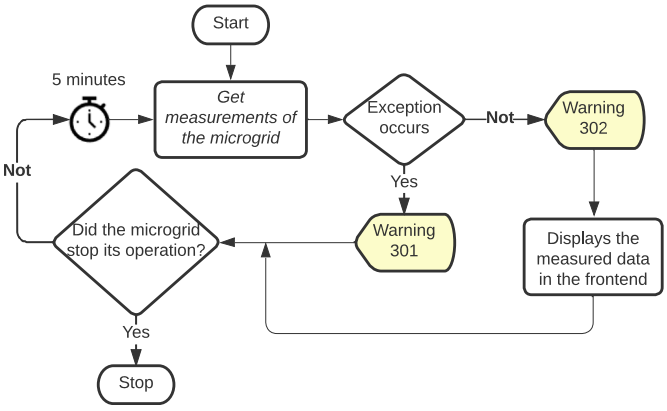



**Fig. 7.** Acquisition and processing of microgrid data.

**Table 2**
Warning messages during the execution of the EMS.

| Code | Warning description |
|---|---|
| 301 | The EMS is not connected to the external API |
| 302 | Node and branch measurements were inserted in the database |
| 303 | The default solution was applied |
| 304 | The optimal solution was applied |
| 305 | There is no communication with the internal API |
| 306 | Measurements older than 24 h have been deleted |

Similarly to real systems, real-time simulations are also subject to communication issues. For example, in case of an internet outage, the communication between the microgrid and the IoT-based EMS is lost. However, as the HTTP requests repeat every 5 min, once the internet connection returns, the data will be collected, stored, and displayed in the frontend.

On the other hand, every 24 h, the algorithm in Fig. 8 is executed. First, a request is made to the EMS API to obtain information about

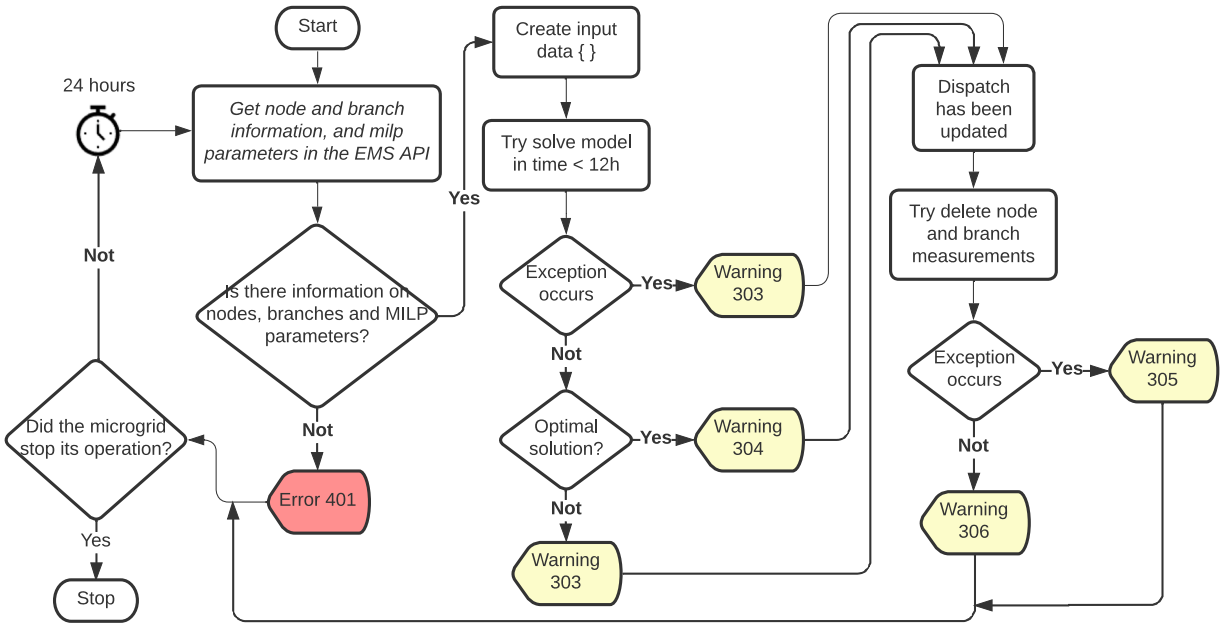



**Fig. 8.** Definition of the optimal day-ahead dispatch.

**Table 3**
Error messages during the execution of the EMS.

| Code | Error description |
|---|---|
| 401 | Nodes, branches, and MILP parameters data are empty |

nodes, branches, and MILP parameters. Information about the topology of the microgrid is essential to calculate the economic dispatch. Thus, if there is no system data, error 401 will appear (see Table 3), and the process will be idle for the next 24 h. Otherwise, an input data dictionary is created, and the MILP model is solved in a time limit of 12 h. If the model is not solved within the time limit, warning 303 will be displayed. Otherwise, the optimality of the solution is analyzed. If the solution is optimal, warning 304 will be shown; otherwise, warning 303 will be shown. Finally, the optimal or default dispatch is updated in the database and the frontend.

The default dispatch assumes that the microgrid operates in grid-connected mode, but it guarantees that at least 25% of the BESS will be available throughout the day in case of a contingency. The remaining 75% capacity is used to dispatch the BESS in a way in which it is fully charged in the afternoon and discharged in the evening.

The historical EMS data contains 24 h of operation, and to prevent overloading the database, a cleaning routine is implemented and executed every 24 h, as shown in Fig. 8. If it is impossible to delete the measurements, warning 305 will be displayed; otherwise, warning 306 will be shown. Finally, the entire process is repeated every 24 h.

## 4. Case study and results

### 4.1. Test setup

The proposed IoT-based EMS is validated using data from a real unbalanced three-phase AC microgrid being deployed at the UNICAMP, in São Paulo, Brazil. A real picture of the CAMPUSGRID microgrid is shown in Fig. 9. Note that some buildings from the university are integrated into the microgrid, including a gymnasium with a PV generation system of 336 kWp, additional 400 kWp PV generation will be installed (totaling 736 kWp), a 150 kVA thermal generator (genset), and an 810 kW/810 kWh BESS [47]. The nominal voltage of the microgrid




**Fig. 9.** DERs in the CAMPUSGRID microgrid, located at the State University of Campinas (UNICAMP), in São Paulo, Brazil [47].

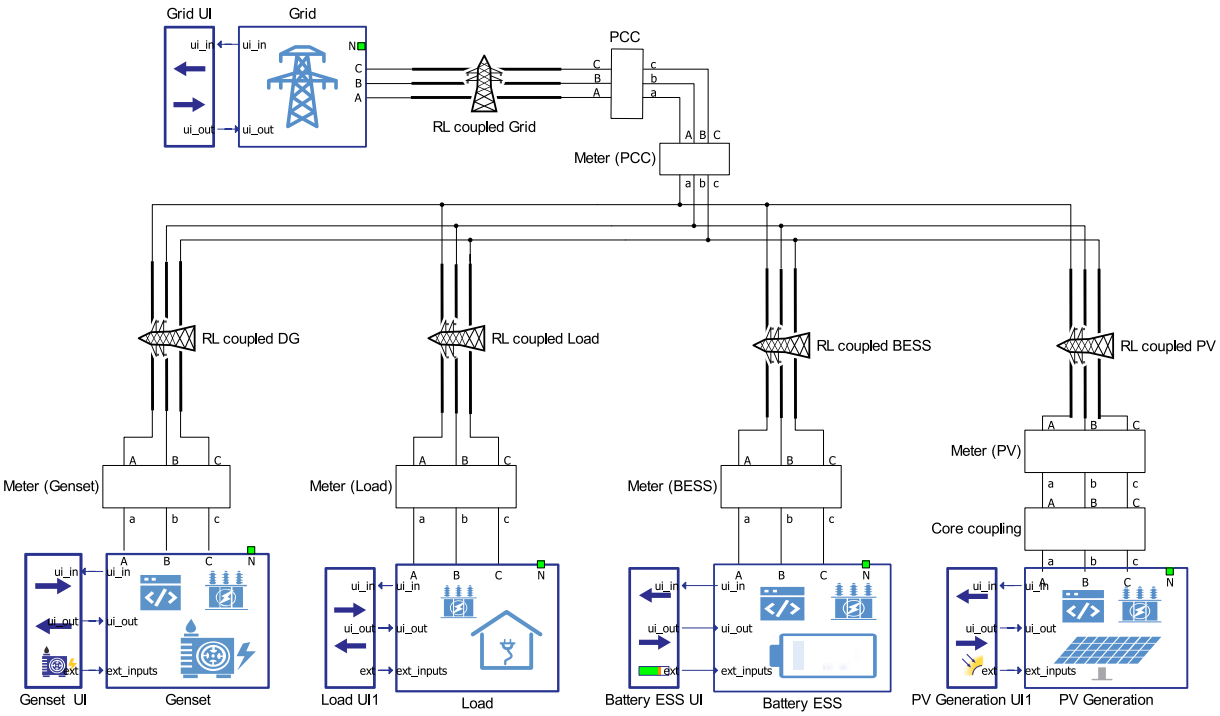
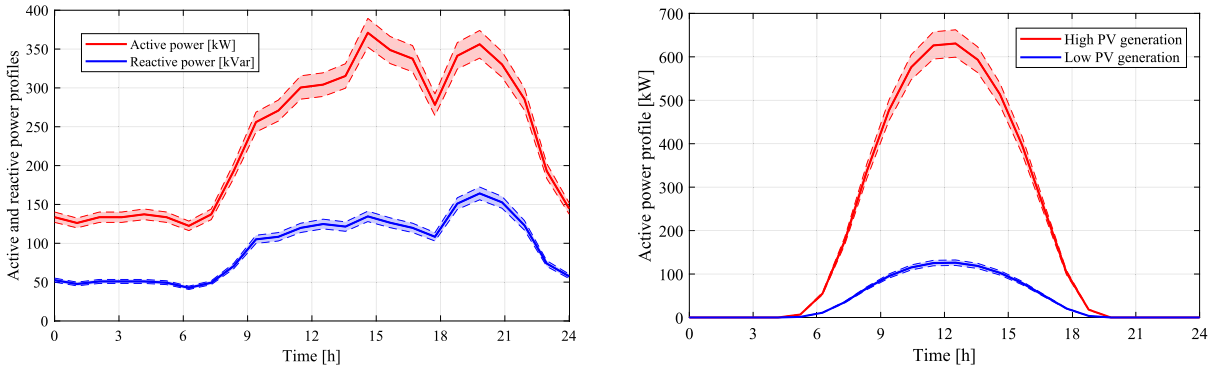



**Fig. 10.** Schematic of the CAMPUSGRID microgrid in Typhoon HIL.




(a) Active and reactive power demands (b) Active power PV generation profile

**Fig. 11.** Average profiles of the CAMPUSGRID microgrid considering a forecast error of 5%.

is 11.9 kV. The PCC connects the microgrid with the main feeder, whose location is also shown in Fig. 9.

Fig. 10 shows the CAMPUSGRID modeled in the schematic editor of the Typhoon HIL [24]. Note that the load and PV generation represent the net demand and PV generation within the microgrid. The sum of the individual loads provides a peak demand of nearly 371 kW, whereas the total capacity of the PV generation is 736 kWp. The active and reactive power demand and PV generation associated with the CAMPUSGRID microgrid are shown in Fig. 11, with a depicted forecasting error of ±5%. The high and low PV generation profiles presented in Fig. 11 represent a sunny and cloudy day, respectively, and are utilized in the case studies. Additionally, the three-phase impedances of the distribution network were considered in the model.

The experimental setup in Fig. 12 shows the Typhoon HIL604 that emulates the real-time operation of the microgrid, connected to a computer via USB. This computer contains the SIL-API application with the Microgrid Config, the Data Logger, and the API. Moreover, two notebooks are used to showcase the proposed IoT-based EMS frontend and the EMS API (i.e., the backend) in execution, communicated via Internet.

To evaluate the effectiveness of the proposed IoT-based EMS considering grid-connected, islanded, and transition modes, the following cases are presented:

- Case I: Without contingencies and high PV generation profile;
- Case II: Without contingencies and low PV generation profile;
- Case III: With contingencies and high PV generation profile;
- Case IV: With contingencies and low PV generation profile;

Note that the high PV generation profile represents a sunny summer day, and the low PV generation profile represents a cloudy winter day. Both with similar forecasting errors of ±5%.

### 4.2. Optimal microgrid operation

For each case mentioned above, the optimal dispatch was determined by the stochastic EDO, based on the characteristics of the microgrid and the parameters, for instance, the hourly energy prices defined by the user in the Energy Prices tab of the EMS (see Fig. 13). In these cases, there was no dispatch of the genset, disconnection of the

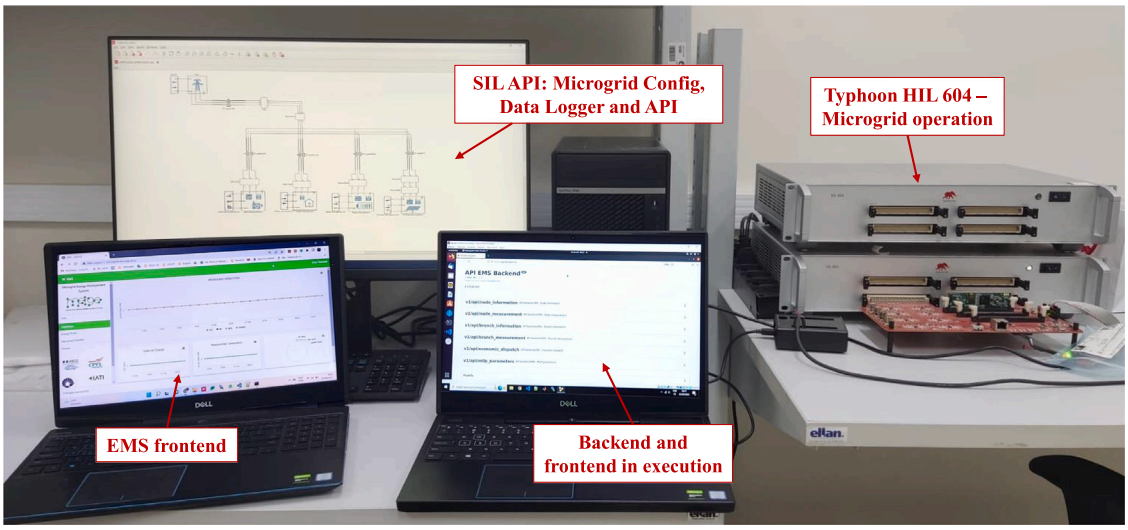



**Fig. 12.** Test setup of the simulation.

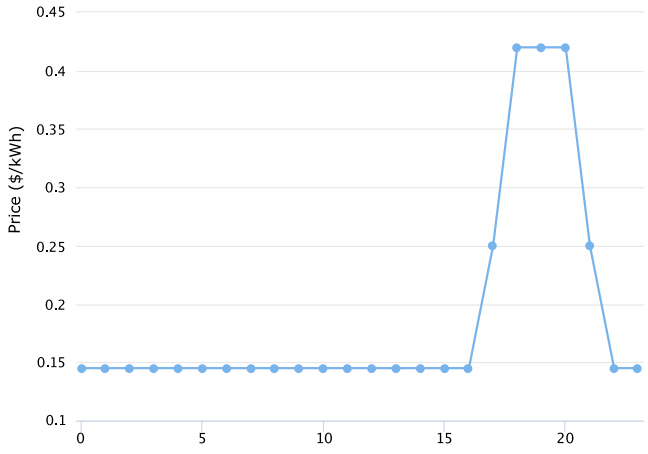



**Fig. 13.** Hourly energy prices ($/kWh).

PV generation, and no load curtailment. Therefore, Fig. 14 presents the scheduled dispatches for the BESS for all study cases. Additionally, the optimization process performed by the EDO is executed only once every 24 h for each case, and the solution was obtained in approximately 10 s.

Case I considers only grid-connected operation, for this reason, the BESS charging takes place during the day when there is solar energy production, and the energy cost is lower. On the other hand, in case II, since the PV generation is low, part of the BESS charging takes place at dawn, when the demand and energy costs are lower. Case III and IV consider a contingency that occurs at 11:00 h, with two hours of duration. During the outage, the BESS operates as grid forming, i.e., controlling the voltage and frequency of the system and being the reference of the grid for the other DERs of the microgrid. In case III, the BESS charging occurs during the islanded operation due to the high PV generation, unlike case IV, where the BESS is discharged. Additionally, in all cases, the BESS discharges during the peak period.

The operation of the CAMPUSGRID microgrid for each case is shown in Fig. 15 (i.e., images captured from the frontend) with an operation horizon of 24 h, starting at 22:00 h until 21:00 h, the next day. The measurements are collected and stored in the database every 5 min. Instantaneous measurements can present outliers. Thus, an average of the last ten measurements is used to avoid displaying values that do not represent the steady-state operation of the microgrid at that specific hour. Then, the average value is updated in the graph using a rolling window. Note that the microgrid controllers follow the scheduled dispatch by the IoT-based EMS. Due to the high operating cost of the genset, it is never dispatched (orange line). The peak of PV generation in cases I and III reached values greater than 500 kWp, unlike cases II and IV where it only reaches 250 kWp. During grid-connected operations, the excess PV generation is injected into the main grid in cases I and III. For this reason, the power at the PCC has negative values.

Due to the averaging effect, it is possible to observe a small amount of power at the PCC after the contingency (i.e., outage from 11:00 h to 13:00 h, Cases III and IV). However, this value decreases until it eventually reaches zero. The excess of PV generation is used to charge the BESS. In contrast, despite not having an excess of PV generation in cases II and IV, the BESS is charged when the energy costs are low. Additionally, during the outage in case IV, the BESS was discharged to meet the demand and to avoid having high costs due to the genset operation or load curtailment. Finally, note that the BESS is discharged in the peak period (18:00 h to 21:00 h), contributing to cost reduction in the microgrid's operation.

Fig. 16 shows the energy of the BESS during the microgrid operation. The minimum SOC was set as 15%, so the minimum energy is 1215 kWh. Note that the BESS starts the day at its minimum energy. In cases I and III (with high PV generation), the energy curve is similar: the BESS charges throughout the day and discharges during the peak period. On the other hand, in case II, as the PV generation is low, the BESS starts charging at the beginning of the day (02:00 h) and reaches its maximum of 810 kWh at 13:00 h. Differently, in case IV, the BESS charges and discharges twice to supply the demand during the outage and reduce energy costs during the peak period.

Table 4 compares four total daily consumptions (i.e., energy from the grid), operation costs, and generations. Through Table 4, it is possible to notice the significant impact of PV generation on a microgrid operation. For instance, cases II and IV (with low PV generation) present total daily consumption and operation costs larger than cases I and III. Note that, when analyzing cases with low PV generation (i.e., cases II and IV), case IV (with contingency) has a higher operational cost. This is due to the fact that there was a greater consumption for charging the battery twice that day. On the other hand, case III presents the lowest total daily consumption and operating cost of the microgrid due to high PV generation. Additionally, when comparing case III with I, in a few moments, part of the power used to charge the BESS comes from the PCC and not only from the excess of PV in case I.

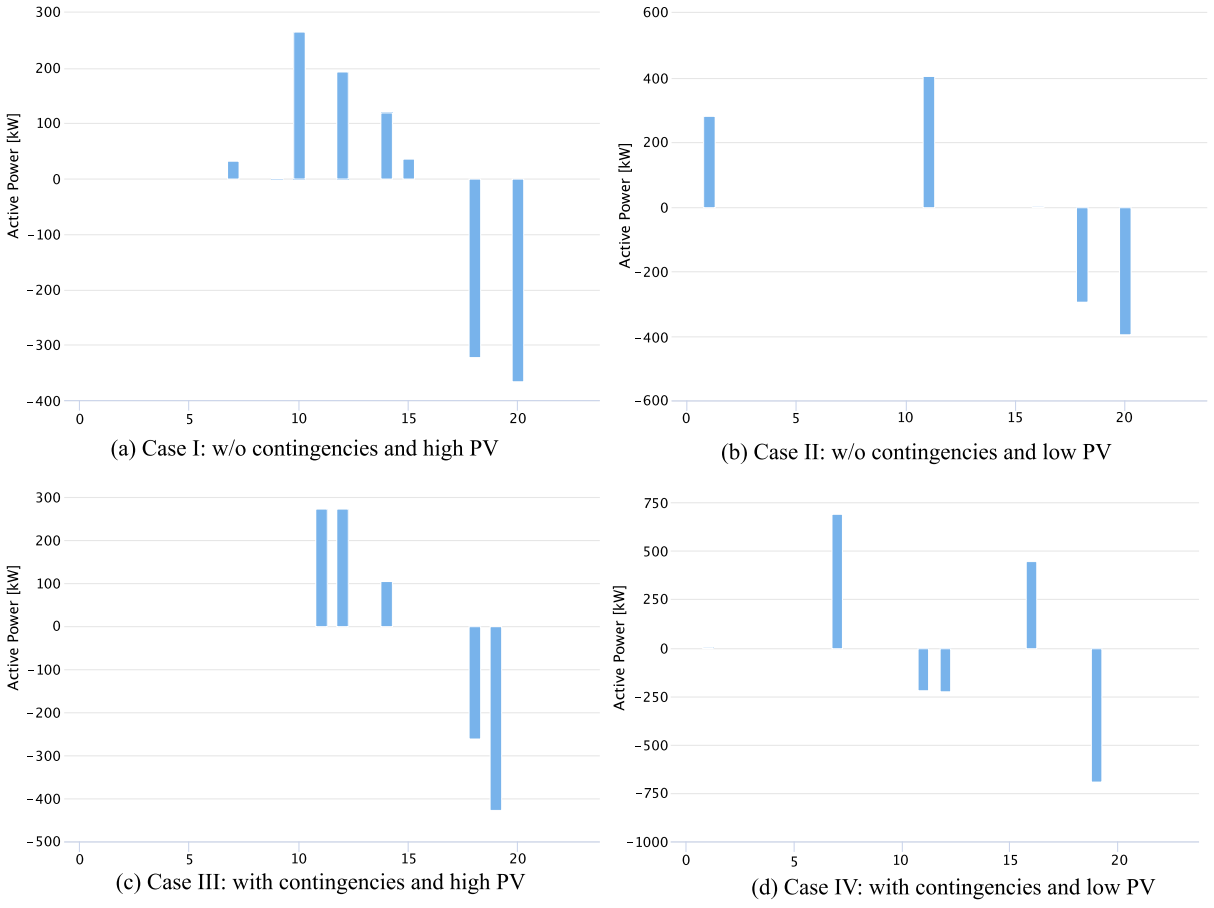



**Fig. 14.** BESS dispatch defined by the EDO for each study case.

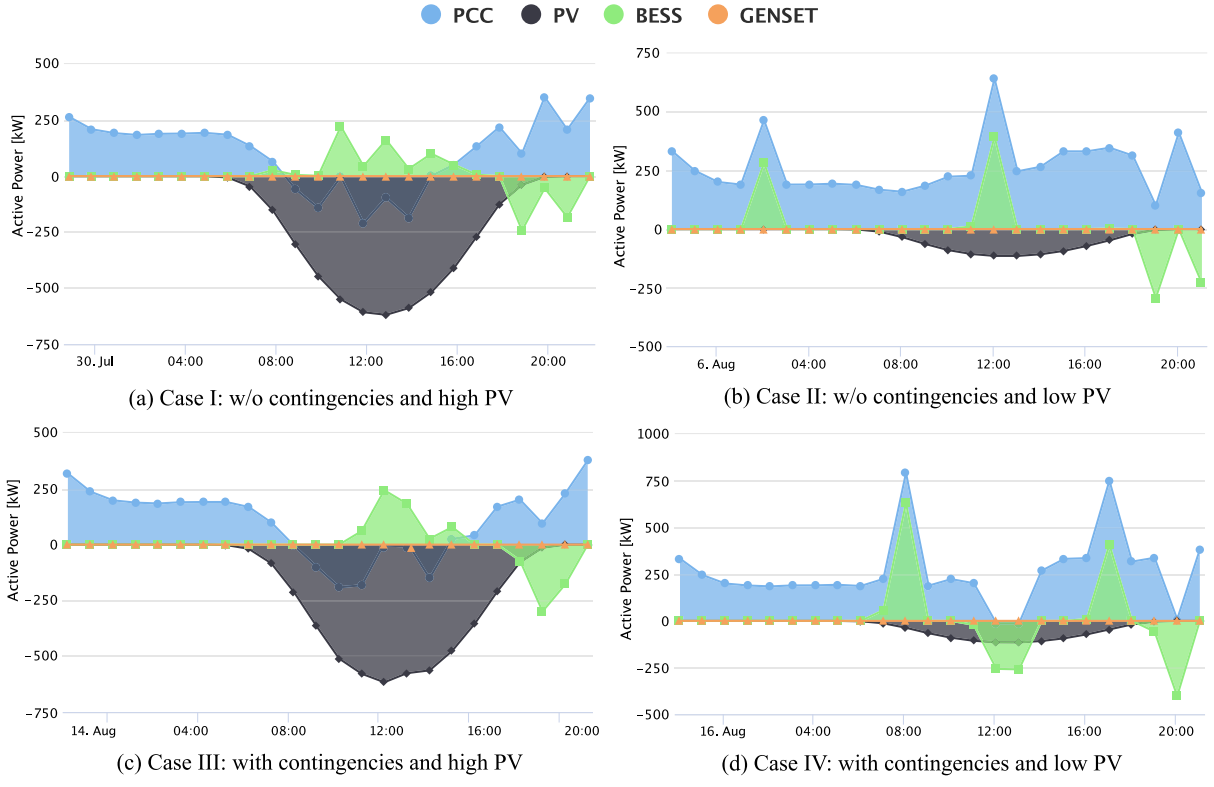



**Fig. 15.** Microgrid 24-hour operation for each study case.

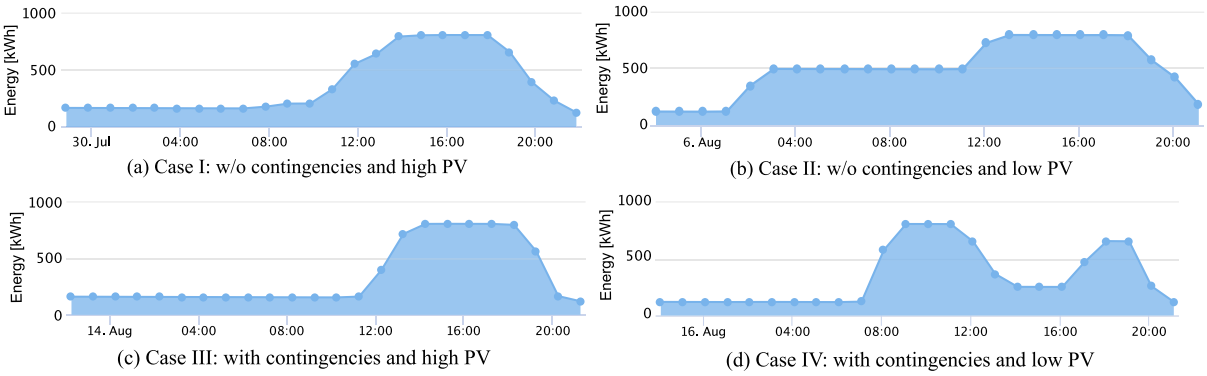



**Fig. 16.** Energy of the BESS for each study case.

**Table 4**
Comparison of total daily consumption (i.e., energy from the grid), operation costs, and generation after 24 h of operation.

| Cases | Total daily consumption [kWh] | Total daily operation cost [BRL] | Total daily generation [kWh] |
|---|---|---|---|
| I | 2512.13 | 601.62 | 5144.36 |
| II | 6312.42 | 1171.83 | 1577.51 |
| III | 2432.37 | 557.70 | 5241.40 |
| IV | 6255.39 | 1175.31 | 1991.57 |

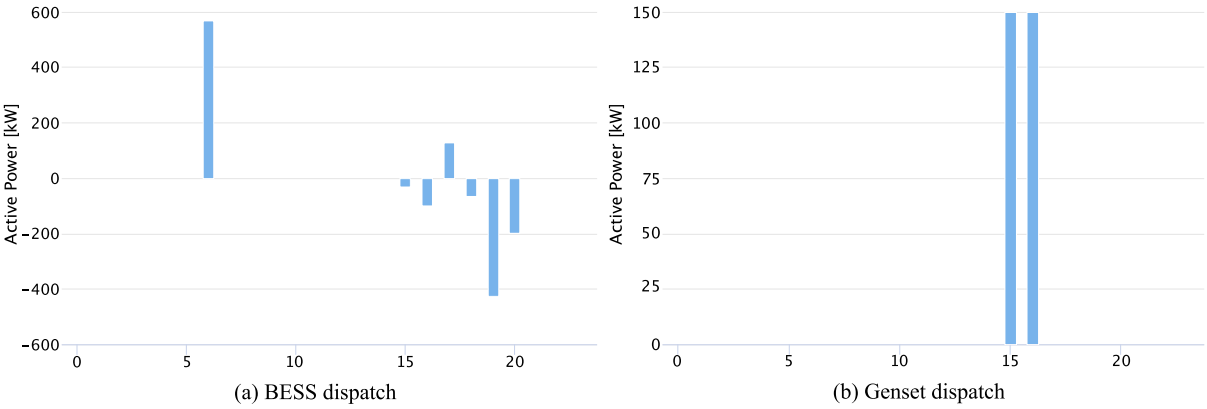



**Fig. 17.** BESS and genset dispatch defined by the EDO.

### 4.3. Tests considering the genset operation and PV curtailment

In order to visualize the genset operation and PV system curtailment in the IoT-based EMS, two new studies cases are presented. To visualize the genset operation, an outage from 15:00 h until 17:00 h is simulated, and a scenario with an average PV generation profile is used. Fig. 17 shows the day-ahead dispatch defined by the EDO for the BESS and the genset operation. Note that the genset is only dispatched during the contingency due to its high operational costs. On the other hand, to visualize the PV system curtailment, an outage from 12:00 h until 14:00 h is simulated, and a scenario with a high PV generation profile is used. Fig. 18 shows the day-ahead dispatch defined by the EDO for the BESS and the PV system curtailment.

Fig. 19a shows the microgrid operation during the 24 h, starting at 21:00 h until 20:00 h. At the beginning of the microgrid operation, the BESS was discharging, i.e., it started the day-ahead at its minimum SOC and charged during the morning. In this case, there was not an excess of PV generation, and the grid supplied the rest of the demand during the grid-connected operation. In contrast, during the islanded operation, the PV generation, genset, and BESS supply the demand (see Fig. 20a). Likewise, as in cases III and IV, the BESS acts as a grid forming, controlling the voltage and the frequency during the outage. Finally, after the contingency, the BESS charges again and discharges during peak hours.

Fig. 19b shows the microgrid operation during the 24 h, starting at 08:00 h until 07:00 h. As the PV generation is greater than the demand, and the BESS was fully charged at the beginning of the contingency, it was necessary to deploy the PV system curtailment to guarantee the internal power balance. At that moment, the BESS supplied the demand (Fig. 20b). Then, the PV generation supplies the demand, and its excess is utilized to recharge the BESS. After the contingency, during the grid-connected operation, BESS is fully charged in preparation for peak hours, where it is again discharged.

Finally, Fig. 21 shows the energy of the BESS during the microgrid operation for the cases that consider the genset operation and PV curtailment. In Fig. 21a, the BESS reaches its minimum energy at 22:00 h, and its maximum in the morning, preparing for a possible outage. Then, in Fig. 21b, the BESS starts its operation fully charged. During the outage, the BESS discharges. However, it reaches its minimum energy during peak hours.

### 4.4. Grid impact

To analyze the voltage and loading profiles, two new simulations were carried out considering a high PV generation profile and a low

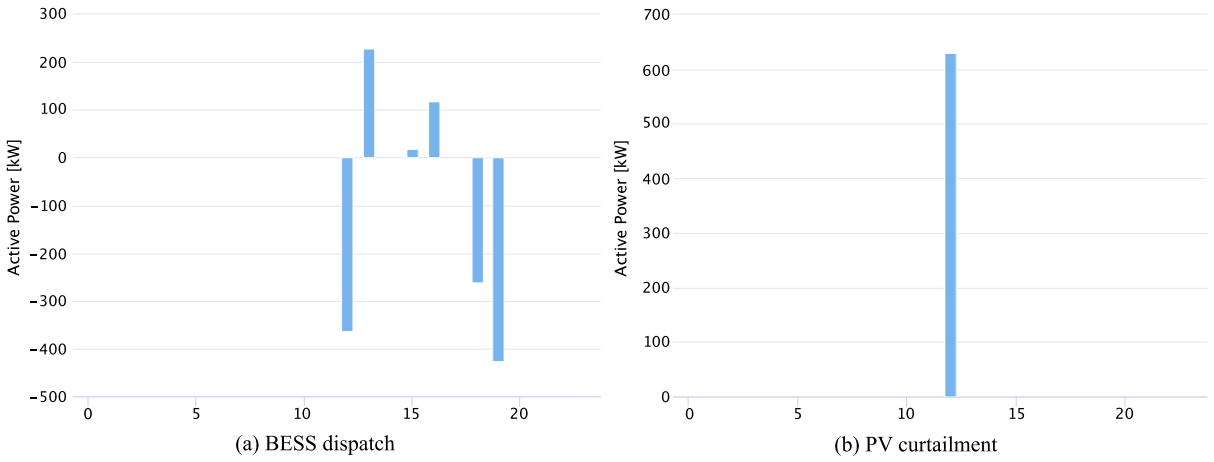



**Fig. 18.** BESS dispatch and PV curtailment defined by the EDO.

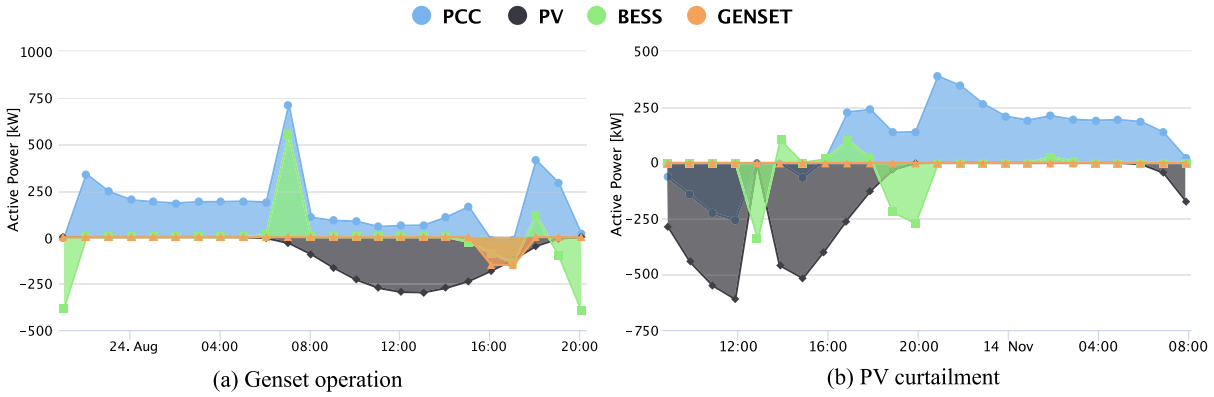



**Fig. 19.** Microgrid operation considering genset dispatch and PV curtailment for different days.

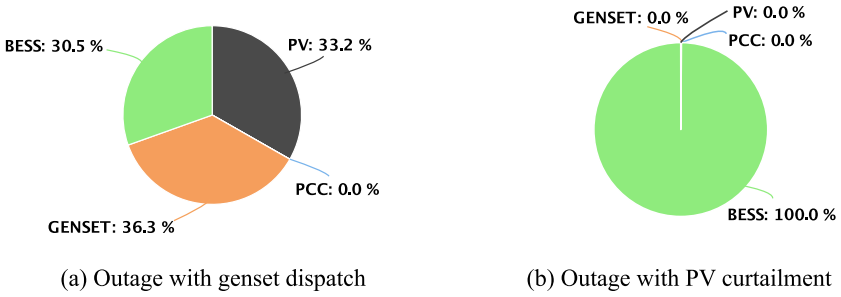



**Fig. 20.** Generation sources at 16:30 h (a) and 12:30 h (b), during the islanded operation.

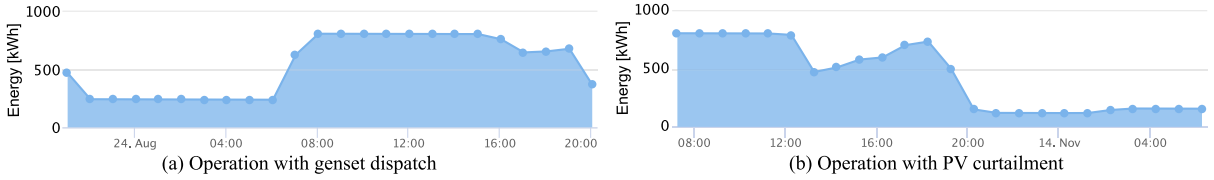
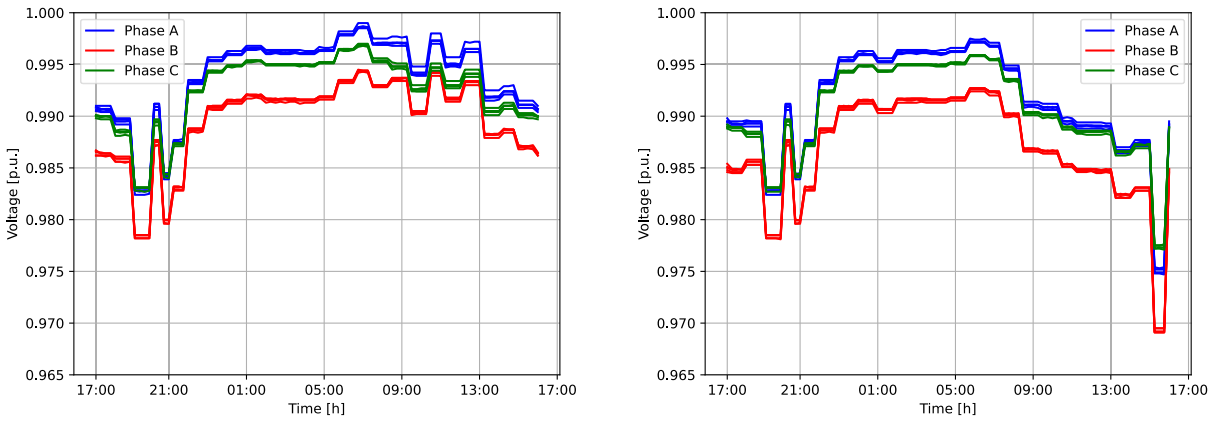



**Fig. 21.** Energy of the BESS during the microgrid operation considering genset dispatch and PV curtailment.

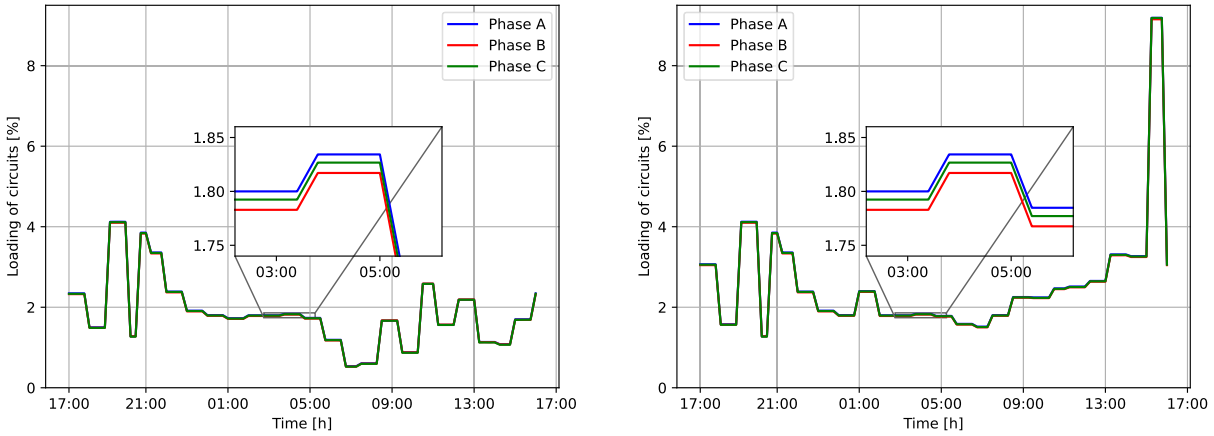
PV generation profile with the same demand as in the previous cases, see Fig. 11(b). The voltage profiles per unit for all phases and nodes of the microgrid are shown in Fig. 22.

As observed, the voltage values are between 0.968 p.u. and 0.998 p.u., maintaining the voltage limits for a medium-voltage network. Moreover, it is possible to note a clear imbalance among phases, which phase B is the most imbalanced. Voltage imbalance occurs due to the characteristics of the three-phase impedances of the microgrid's internal and external network. In this case, the voltage imbalance is mostly caused by the unbalanced nature of the equivalent model used for the main grid. In Figs. 22(a) and 22(b), an abrupt drop and increase in the voltage can be seen due to the BESS charging and discharging.




(a) High PV (b) Low PV

**Fig. 22.** Voltage magnitude profile of the microgrid for all phases and nodes.




(a) High PV (b) Low PV

**Fig. 23.** Loading of the circuit of the microgrid for all phases and branches.

There is an increase in the voltage profile during sunny hours (07:00 h to 17:00 h) for the case with high PV generation compared to the low PV generation scenario. Additionally, the voltage drops during the afternoon due to the increased demand.

Fig. 23 shows the loading of circuits for each branch and phase of the microgrid, considering a high and low PV generation profile. The maximum value is about 4,1% of the circuit capacity in the case with high PV generation and 9,2% in the case with low PV generation. Similar to Fig. 22, in Fig. 23(a) and Fig. 23(b), there are abrupt variations in the loading profile of the lines due to the BESS charging and discharging.

## 5. Conclusions

This paper proposes a new energy management system (EMS) based on the Internet-of-Things (IoT) for the optimal operation of unbalanced three-phase AC microgrids. The platform provides for the optimal day-ahead scheduling of distributed energy resources (DERs) and considers direct load and photovoltaic control. Additionally, unexpected islanded operation at any time is considered via contingency constraints. Uncertainties associated with demands and photovoltaic (PV) generation are considered via stochastic economic dispatch optimization (EDO). The IoT-based EMS monitors the microgrid operation in real-time, exhibiting the data operation in a graphical user interface (GUI). Tests are performed using real data from the CAMPUSGRID microgrid, which is being developed at the UNICAMP, via real-time simulations utilizing Typhoon HIL.

Results showed that the tool is able to define the day-ahead dispatch of DERs and monitor a microgrid in real-time, under realistic conditions in which communication, optimality, and stability issues are taken into account and humans are interacting with the system. The high production of PV energy significantly impacts the reduction of microgrid operating costs. It was observed that in cases with high PV generation, the renewable energy resources (RESs) are enough to supply the conventional demand and to charge the battery energy storage system (BESS) during islanded operation. Furthermore, the genset is only required during outages, as the last option among the RESs to be dispatched due to its high operational cost. In most cases, the BESS charges throughout the day, taking advantage of the low energy price and the high PV generation, and it discharges during the peak period. Results also showed that operational costs are higher when islanded operation and low PV generation are considered simultaneously. During an outage at noon, with a fully-charged BESS and high PV generation, PV curtailment was necessary, and the BESS supplied the demand. However, load curtailment was always avoided. Finally, as the platform is based on a microservice architecture, new modules with different functionalities can be added. Thus, future works will add an improved forecasting tool, a predictive control model, and a stability assessment module to the IoT-based EMS.

## CRediT authorship contribution statement

**Jéssica Alice A. Silva:** Writing – original draft, Conceptualization, Software, Methodology, Visualization. **Juan Camilo López:** Methodology, Conceptualization, Supervision, Writing – review & editing. **Cindy Paola Guzman:** Conceptualization, Supervision, Writing – review & editing. **Nataly Bañol Arias:** Conceptualization, Supervision, Writing – review & editing. **Marcos J. Rider:** Supervision, Writing – review & editing, Project administration. **Luiz C.P. da Silva:** Supervision, Writing – review & editing, Project administration.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jessica Alice Alves da Silva reports financial support was provided by Brazilian institution São Paulo Research Foundation. Jessica Alice Alves da Silva reports financial support was provided by CPFL Energy.

## Data availability

The authors are unable or have chosen not to specify which data has been used.

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